UNIVERSITY OF KELANIYA – SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science – Management & Information Technology (Honours) Degree Examination – January / February 2024

Academic Year 2021/2022 – Semester II

Information Technology
INTE 22212 – Software Design Patterns and Frameworks

No. of Questions:4 **No of Pages:3** **Time: 2 1/2 hrs.**

*Answer **ALL** questions. When allocating time for questions, take the marks allocated into consideration. State any assumption that you make.*

*This is an **OPEN BOOK** exam.*

QUESTION 1

(25 Marks)

The salary calculation has various components as shown below.

Over Time = Basic Salary * Number of Overt Time Hours / (30 * 8)

Attendance Allowance = If Absent days > 3, No Allowance; IF the absent days > 1 and < 3, 50% of the Basic Salary, if Absent days =0, 75% of the Basic salary

Gross salary = Basic Salary + Attendance Allowances + Over Time

EPF Employee contribution = Gross Salary * 0.08

EPF Employer contribution = Gross Salary * 0.12

ETF = Gross Salary * 0.03

No Pay = Basic Salary * No Pay Days / 30

Tax = (Gross Salary – 100,000) * 0.06

Total Deduction = EPF Employee Contribution + No Pay + Tax

Net Salary = Gross Salary – Total Deduction

Total EPF = EPF Employee contribution + EPF Employer contribution

In this organization, salaries of permanent employees will be calculated according to the above calculations; while there are few deviations for other employees as Executive employees, there is no overtime and no pay.

1.1 Draw a diagram to indicate the above scenario. **[4 Marks]**

1.2 What design pattern can be employed for sub-systems that calculate different salary components for the different employees? Explain the reasons for selection. **[5 Marks]**

1.3 What challenges will you face when you design the system with the selected design pattern? **[4 Marks]**

1.4 Explain the mechanism that you will use to overcome those challenges. **[4 Marks]**

1.5 Many of the parameters can be changed over time. How would you improve the configurability of the entire system? State the configuration [**4 Marks]**

1.6 At the time of salary calculation, no other user should be logged into the system and while the calculation process is executing, no other processes should be processed. What is the design pattern that you propose and why? **[4 Marks]**

QUESTION 2

(25 Marks)

A Supermarket has functions such as purchasing, sales, warehousing etc. You are hired to design a system to facilitate the supermarket system. This supermarket is part of a supermarket chain.

2.1 Draw a diagram to depict the above scenario. **[4 Marks]**

2.2 How can an organization effectively implement and manage a micro services architecture to achieve scalability, fault isolation, and ease of continuous deployment? **[5 Marks]**

2.3 Propose micro-service architecture for the above problem. **[4 Marks]**

2.4 What are the methods that you can propose to maintain the data consistency in the proposed micro service architecture? **[4 Marks]**

2.5 How would you achieve security in the above proposed micro-service architecture for the above problem? **[4 Marks]**

2.6 Why is it essential to provide the same hardware configurations and options for each micro service platform? **[4 Marks]**

QUESTION 3

(25 Marks)

You are hired to design a system for a restaurant. This restaurant has different flavors of coffee. For example, Irish Coffee is a cocktail consisting of hot coffee, Irish whiskey, and sugar, topped with cream, while Nitro Coffee is a cold brew coffee infused with nitrogen gas, served on tap. It has a creamy texture and is often compared to stout beer.

3.1 Considering the many combinations that exist for coffee, what is the proposed design pattern? **[5 Marks]**

3.2 Draw a simple class diagram to facilitate the above design pattern. **[5 Marks]**

3.3 The restaurant needs to connect to different websites to show different events that are organized by different entities. Considering that these websites are developed by different technologies by different vendors, what is your solution to integrate them? **[5 Marks]**

3.4 If you are required to implement logging mechanism, what is the design pattern you can utilize. Provide logical reasons for your selection. **[5 Marks]**

3.5 How can you extend this design to achieve High Availability (HA) and Disaster Recovery (DR)? **[5 Marks]**

QUESTION 4

(25 Marks)

Explain the possible design pattern you can suggest for the following scenarios. Provide any simple class diagrams.

4.1 You are designing a system for a fantasy cricket team. The user should be able to select a maximum of five batsmen, a maximum of five bowlers and one wicketkeeper. There should be one captain. How would you use an extended singleton design pattern to achieve the above tasks? **[5 Marks]**

4.2 To design a banking system with different types of accounts, such as Current. Saving and Children accounts. **[5 Marks]**

4.3 An exception-handling system that can handle different priorities depending on the problem. **[5 Marks]**

4.4 Developing a Learning Management System (LMS), and when users read a large content, it is required to load only a couple of pages of the documents rather than the entire document. **[5 Marks]**

4.5 When you need to develop a website with different authorization options, it depends on the user's need. **[5 Marks]**